

INFORMATION BROCHURE

For Admission to

**Ph.D. Programme: Full-Time (with Scholarship)/
Part-Time/Self-Financed/Sponsored
AY: 2025-26 (July Session)**



NATIONAL INSTITUTE OF TECHNOLOGY GOA

Cuncolim, South Goa, GOA – 403703

www.nitgoa.ac.in



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1. Important Information

Applications are invited for the admission to Ph.D. programmes (Full-Time (with scholarship)/Part-Time/Self-Financed/Sponsored) for the academic year AY: 2025-26 (July Session) at the following Departments of NIT Goa:

- Department of Computer Science and Engineering (CSE)
- Department of Electrical and Electronics Engineering (EEE)
- Department of Electronics and Communications Engineering (ECE)
- Department of Civil Engineering (CVE)
- Department of Mechanical Engineering (MCE)
- Department of Applied Sciences (APS)
- Department of Humanities and Social Sciences (HSS)

Applicants must apply through Online Portal: <https://www.nitgoa.ac.in/admissions/phd.html>

A student applying for more than one Department will have to fill separate application forms.

Applicants are advised to read this Information Brochure carefully and Check the Reservation Chart (for Full-Time with scholarship ONLY) in page 7 before applying online.

Admission to Ph.D. programmes will be in the following schemes:

✓ **Full-Time (with scholarship):**

The candidates selected under this scheme shall be entitled for scholarship/assistantship and House Rent Allowance (HRA) as per Govt. of India norms from time to time. Please read the scholarship requirement section for detailed information.

✓ **Part-Time**

Part-time Ph. D programme is offered for working professionals from Govt. R&D Organizations /Public Sector /Industries /National Research Laboratories and for faculty members from Govt./Govt. Aided /Private Engineering/Science and Arts Colleges/Universities. It is necessary for a candidate applying under part-time scheme to produce a “**No Objection Certificate**” on a proper letter head from the appropriate authority in the organization where he/she is presently employed. Students admitted under part-time scheme have to serve a minimum one semester residential period at the Institute.

However, regular employees of centrally funded research laboratories and organizations, who have a minimum of five years of continuous work experience in the same organization and are selected for the part-time Ph.D. programme, will be permitted to complete their Ph.D. coursework through NPTEL/MOOC platforms. The courses must be pre-approved by the Departmental Research Committee (DRC), and such candidates shall not be required to be physically present for attending classes.

✓ **Self-Financed**

- This is a full-time Ph.D program without scholarship.
- Under this scheme, the candidate has to be physically present in the campus.
- The candidate must fulfill all other full-time Ph.D. program requirements.

✓ **Sponsored:**

- This is a full-time Ph. D program without scholarship.
- The Institute may admit persons who are employed in a Central/State Govt. Departments/ PSUs/ Reputed Educational Institutes/ Research organizations/ Reputed Industries for doing research in the Institute on a full-time basis.
- Candidates may be sponsored by a company or by educational institute or Industry or an Organization or Department etc.
- Under this scheme, the candidate has to be physically present in the campus.
- The candidate must fulfill all other full-time Ph.D. program requirements.

2. Eligibility and Academic Requirements

- **For Ph. D in Engineering and Technology (CSE, ECE, EEE, CVE, MCE):** Minimum 6.5 CGPA on a 10 point scale grading system or 60% marks in both Bachelor's and Master's (M.E. or M.Tech in relevant discipline) level from recognized Technical Institute or University. For SC/ST candidates a minimum 6.0 CGPA on a 10 point scale grading system or 55% marks.

Exceptional candidates (with CGPA ≥ 9.0) with B.Tech./B.E from a reputed Institute are also eligible to apply for the Ph.D. programme.

- **For Ph. D in Applied Sciences:** Minimum 6.5 CGPA on a 10 point scale grading system or 60% marks in both Bachelor's and Master's (Master's degree in Sciences / M. Tech. in relevant discipline) level from recognized Institute or University. For SC/ST candidates a minimum 6.0 CGPA on a 10 point scale grading system or 55% marks.
- **For Ph. D in Humanities & Social Sciences:** Minimum 6.0 CGPA on a 10 point scale grading system or 55% marks in both Bachelor's and Master's (Humanities/Social Sciences) level from recognized Institute or University. For SC/ST candidates a minimum 5.5 CGPA on a 10 point scale grading system or 50% marks.

3. Indicative list of Broad Areas of Research

❖ Computer Science and Engineering (CSE)

- **Full-time (with scholarship):** Brain-Computer Interface, Machine Learning, Computer Vision, Internet of Things, Cyber Physical Systems, Natural Language Processing, AI for Agriculture, Applied machine learning for speech/audio, Vision, Text and Time series data, Audio deep fake detection, Data Mining, Machine Learning, Social Media Analysis, Wireless Sensor Networks, Internet of Things, Swarm Intelligence, Optimization Algorithms, High Performance Computing, Quantum Computing, AI/ML for communication network: Federated learning, IoT-Edge- Cloud Continuum Systems, Advanced Mobile Communication (5G/6G), Game theory application in advanced computer network, Soft Computing, Blockchain, Adversarial Machine Learning, Cyber Security.
- **Part-Time/Self-Financed/Sponsored:** Brain-Computer Interface, Machine Learning, Computer Vision, Internet of Things, Cyber Physical Systems, Natural Language Processing, AI for Agriculture, Applied machine learning for

speech/audio, Vision, Text and Time series data, Audio deep fake detection, Machine Learning, Social Media Analysis, Data Mining, Wireless Sensor Networks, Internet of Things, Swarm Intelligence, Optimization Algorithms, High Performance Computing, Quantum Computing, AI/ML for communication network: Federated learning, IoT-Edge- Cloud Continuum Systems, Advanced Mobile Communication (B5G/6G), Game theory application in advanced computer network, Soft Computing, Blockchain, Adversarial Machine Learning, Cyber Security.

❖ **Electrical & Electronics Engineering (EEE)**

- **Full-time (with scholarship):** Smart Electric Grid, Electric vehicles, Wireless Power Transfer, EV Charging topologies, Battery Management System, Grid connected/Stand-Alone PV Systems, Renewable Energy Systems, Optimization Techniques, Power Quality Issues, HVDC and FACTS, Applications of Soft Computing Techniques, Restructured Power Systems; Electric Arc Furnace Operations; AC/DC Microgrids, Multilevel Converter, Pulse width Modulation, Switched Reluctance Motor and Drives, Digital Signal/Image Processing, Design of Wavelets, FPGA based Hardware Accelerators, Biometrics and other relevant research domains/areas of Electrical and Electronics Engineering.
- **Part-Time/Self-Financed/Sponsored:** Smart Electric Grid, Electric vehicles, Wireless Power Transfer, EV Charging topologies, Battery Management System, Grid connected/Stand-Alone PV Systems, Renewable Energy Systems, Optimization Techniques, Power Quality Issues, HVDC and FACTS, Applications of Soft Computing Techniques, Restructured Power Systems; Electric Arc Furnace Operations; AC/DC Microgrids, Multilevel Converter, Pulse width Modulation, Switched Reluctance Motor and Drives, Digital Signal/Image Processing, Design of Wavelets, FPGA based Hardware Accelerators, Biometrics and other relevant research domains/areas of Electrical and Electronics Engineering.

❖ **Electronics & Communication Engineering (ECE)**

- **Full-time (with scholarship):** Microelectronics, Digital and Analog VLSI, Network on CHIP. Low power Active filter circuit design. System Design and implementation using an FPGA. Image Denoising, Biomedical image Analysis, Image Classification, Object Tracking, Ultrasound signal analysis, FPGA Based Image Processing Applications, Machine Learning, Deep Fake Detection. VLSI Design, Analog and Mixed Signal Design, AI assisted VLSI circuits. Microstrip Antenna Design; dielectric resonator antenna (DRA) design; Terahertz (THz) antennas; Smart Antennas, Antenna array design for 5G MM-Wave; Adaptive Beamforming; Power Divider Network for Array; Fractal Antenna; Phased Array Antenna; Wearable and Conformal Arrays; Array Pattern Synthesis; Evolutionary Algorithms. Transmitter/receiver design, Fast Adder, Multiplier, multiply-divide unit, Signal processing for wireless communication, Use of machine learning in advanced wireless communication. Intelligent reflecting surface. AI and ML for sensors design at nanoscale, nanoscale optical sensors, Low Cost Point of Care Diagnostics, Biomedical Signal Processing, Pattern and AI in Medicine, Infrared Thermography (Thermal Imaging), Agriculture Electronics, Biomedical Instrumentation. Antenna Engineering, RF and Microwaves, AI and ML for Microwaves, Microwave Wireless Power Transfer, Ground Penetrating Radar,

Microwave Imaging, Inverse Problems. Semiconductor Devices, VLSI Circuit Designs, Digital Circuit Design, System Verilog.

- **Part-Time/Self-Financed/Sponsored:** Microelectronics, Digital and Analog VLSI, Network on CHIP. Low power Active filter circuit design. System Design and implementation using an FPGA. Image Denoising, Biomedical image Analysis, Image Classification, Object Tracking, Ultrasound signal analysis, FPGA Based Image Processing Applications, Machine Learning, Deep Fake Detection. VLSI Design, Analog and Mixed Signal Design, AI assisted VLSI circuits. Microstrip Antenna Design; dielectric resonator antenna (DRA) design; Terahertz (THz) antennas; Smart Antennas, Antenna array design for 5G MM-Wave; Adaptive Beamforming; Power Divider Network for Array; Fractal Antenna; Phased Array Antenna; Wearable and Conformal Arrays; Array Pattern Synthesis; Evolutionary Algorithms. Transmitter/receiver design, Fast Adder, Multiplier, multiply-divide unit, Signal processing for wireless communication, Use of machine learning in advanced wireless communication. Intelligent reflecting surface. AI and ML for sensors design at nanoscale, nanoscale optical sensors, Low Cost Point of Care Diagnostics, Biomedical Signal Processing, Pattern and AI in Medicine, Infrared Thermography (Thermal Imaging), Agriculture Electronics, Biomedical Instrumentation. Antenna Engineering, RF and Microwaves, AI and ML for Microwaves, Microwave Wireless Power Transfer, Ground Penetrating Radar, Microwave Imaging, Inverse Problems. Semiconductor Devices, VLSI Circuit Designs, Digital Circuit Design, System Verilog.

❖ Civil Engineering (CVE)

- **Full-time (with scholarship):** Sustainable Geotechnics, Soil behaviour under static/cyclic conditions, Clay mineralogy and microstructural aspects, Reinforced earth, Environmental Geotechnology, Transportation Geotechnics, Sustainable Construction.
- **Part-Time/Self-Financed/Sponsored:** Sustainable Geotechnics, Soil behaviour under static/cyclic conditions, Clay mineralogy and microstructural aspects, Reinforced earth, Environmental Geotechnology, Transportation Geotechnics, Sustainable Construction.

❖ Mechanical Engineering (MCE)

- **Full-time (with scholarship):** Design for manufacturing and Assembly, Applied Ergonomics, Virtual Reality for manufacturing, Additive manufacturing, Numerical and Experimental Fluid Flow and Heat Transfer, Microfluidics, Battery Cooling Technologies, Green Energy, Energy Harvesting and Energy Storage, Design and Optimization of Thermal System, Micro and Nano Heat Transfer, AI and ML in Thermal System
- **Part-Time/Self-Financed/Sponsored:** Reverse Engineering, Product Design for Elderly and Kids, Design for manufacturing and Assembly, Quality and Reliability, Sustainability, Numerical and Experimental Fluid Flow and Heat Transfer, Microfluidics, Battery Cooling Technologies, Green Energy, Energy Harvesting and Energy Storage, Design and Optimization of Thermal System, Micro and Nano Heat Transfer, AI and ML in Thermal System.

❖ Applied Sciences (APS)

➤ Full-time (with scholarship):

Mathematics: Applied Mathematics, Hydrodynamic Stability, Non-Newtonian Fluids, Fluid Instabilities. Artificial Intelligence and Machine Learning, Numerical Optimization and Simulation, Biomedical Imaging, Inverse Problems. Partial Differential Equations, Fractional Differential Equations, Mathematical Biology, Finite Element Methods.

Physics: Photonics: Fiber Optic Sensors; & Materials Science: Microwave Absorption Materials, Thin Film Coatings and Gas Sensors, Energy Storage Devices.

Chemistry: Electron Paramagnetic Resonance (EPR) of Transition Metal Ions and Organic Free Radicals, Electron Spin Relaxation Time And Its Mechanism, Distance Measurement Between Sites In Biomolecules, Application of EPR towards Biology, Chemistry of Lanthanides and Actinides.

➤ Part-Time/Self-Financed/Sponsored:

Mathematics: Applied Mathematics, Hydrodynamic Stability, Non-Newtonian Fluids, Fluid Instabilities. Artificial Intelligence and Machine Learning, Numerical Optimization and Simulation, Biomedical Imaging, Inverse Problems. Partial Differential Equations, Fractional Differential Equations, Mathematical Biology, Finite Element Methods.

Physics: Photonics: Fiber Optic Sensors; & Materials Science: Microwave Absorption Materials, Thin Film Coatings and Gas Sensors, Energy Storage Devices.

Chemistry: Electron Paramagnetic Resonance (EPR) of Transition Metal Ions and Organic Free Radicals, Electron Spin Relaxation Time And Its Mechanism, Distance Measurement Between Sites in Biomolecules, Application of EPR towards Biology, Chemistry of Lanthanides and Actinides

❖ Humanities and Social Sciences (HSS)

➤ Full-time (with scholarship):

English: Post Humanism and Dalit Literature

Economics: R&D, Innovation, Intellectual Fellowship Property Rights and Sustainable Development.

➤ Part-Time/Self-Financed/Sponsored:

English: Post Humanism and Dalit Literature

Economics: R&D, Innovation, Intellectual Fellowship Property Rights and Sustainable Development.

4. Reservation Chart (for Full-time with scholarship only)

OPEN	OPEN-PwD	EWS	EWS-PwD	OBC	OBC-PwD	SC	SC-PwD	ST	ST-PwD
41	2	9	1	27	1	15	1	8	0

Note: The final student intake is subject to availability of Supervisor.

OPEN = Open Category; EWS = Economically weaker Section; OBC = Other Backward Caste (Non Creamy Layer); SC=Scheduled Caste; ST=Scheduled Tribe; PwD: Person with Disability

5. Important Instructions

- Candidates **MUST** specify broad areas of research in the application form.
- **The candidates with INSPIRE, UGC-CSIR Fellowships or any such fellowships can apply under Full-Time category irrespective of vacancies.**
- Institute reserves the right to reject the application(s) if a faculty member is not available to guide the candidate(s) in his/her proposed research area.
- Incomplete application forms and forms submitted after the last date will be rejected.
- Canvassing in any form will lead to rejection of application form.
- The candidates are required to see our Institute website for fee structure and faculty expertise and are also advised to visit our Institute website regularly for the updates.
- The candidates are advised to read the NIT Goa Ph.D Rules and Regulations ([https://www.nitgoa.ac.in/uploads/PhD%20\(Rules%20&%20Regulations\)_2014-15_Jan%202025_Merge.pdf](https://www.nitgoa.ac.in/uploads/PhD%20(Rules%20&%20Regulations)_2014-15_Jan%202025_Merge.pdf)) for detailed programme structure.

6. Selection Procedure

The admission to Ph.D. Programme will be based on the criteria given below:

- a) Overall academic performance and written test conducted by the respective Department.
 - b) Performance in personal interview.
- The candidates shortlisted in the written test will be called for the personal interview.
 - Short-listed candidates for written test and personal interview will be uploaded on the Institute website. No separate intimation will be sent to the applicants.
 - All candidates selected for admission shall be required to deposit the requisite Institute fee and produce necessary original documents at the time of admission.

7. Fee Structure for the year 2025-26

The applicants may refer the Institute fee structure at the below link:

https://www.nitgoa.ac.in/static/fee_structure_23-24_25july2023.pdf

Fee structure may be changed from time to time as per the Institute norms.

8. Scholarship Requirement

A GATE qualification obtained by candidate at some point in time in the respective discipline or any other eligible fellowships such as JRF/UGC/NET/CSIR/DST-INSPIRE etc. in the respective disciplines or qualification in any other national level tests as specified by the norms of GoI from time to time.

9. Facilities and Opportunities for Ph.D scholars

- **Financial support upto INR 50,000 for meritorious scholars to attend International Conferences abroad.**
- Financial support for contingency expenses and attending workshops/conferences within India.
- Exposure to the latest technological trends and developments through various institutional programs and activities.
- Newly inaugurated campus with state-of-the-art facilities.
- Access to selected top-tier research journals and research databases.
- Single room hostel accommodation for Ph. D scholars in the new campus at Cuncolim
- 24 hours access to academic resources including laboratories.

10. How to apply

Applicants must visit: <https://www.nitgoa.ac.in/admissions/phd.html> for Ph.D. application filling and fee payment.

While filling-in the online application, attention must be paid to the following:

- ✓ Applicants are instructed to thoroughly go through “**Ph.D. Information Brochure**”.
- ✓ Further, eligible applicants may apply through **ONLINE** mode only using “**Ph.D. Online Application Portal**”
- ✓ Each application must accompany an application fee of Rs. 500/- per Department for Gen/OBC(NCL)/EWS applicants and Rs. 300/- per Department for SC/ST/PwD applicants. Applications to more than one Department shall be considered only upon payment of the requisite fees for each Department. Application fee is non-refundable.
- ✓ Candidates can pay the application fee online using Debit/Credit Card/UPI/Net Banking etc.
- ✓ The application fee payment is to be done using the link “**Application Fee Payment**”.
- ✓ The proof of the payment is to be kept ready and the same is to be uploaded at the required place while applying online.
- ✓ Fee payment should be made through online mode only, on or before the last date, i.e., **22 June 2025, 5 PM** **30 June 2025, 5 PM**.
- ✓ Applicants **MUST** be ready with the scanned copy of the following while submitting the online application using “**Ph.D. Online Application Portal**”
 - Application fee payment proof
 - Duly filled Application form (**ANNEXURE-I**) along with relevant supporting documents (**ANNEXURE-II**), ALL into one *.pdf file
- ✓ Online application once submitted by the applicants shall be considered final and binding. Requests for making corrections in the online application shall not be entertained once the final submission is made.
- ✓ Candidates are instructed to send the softcopy (PDF) of their finally submitted application (without supporting documents) to phdadmissions@nitgoa.ac.in with mail subject: “**Ph.D. Application July 2025 <Department Name>**”.

11. Supporting Documents

Applicants MUST be ready with the following scanned documents, whichever applicable, while submitting the online Ph.D. application form.

- i. GATE/UGC/CSIR/NBHM/NET qualified certificate. (For Full-Time with scholarship only)
- ii. Master’s Degree/Provisional certificate. If the result of qualifying degree is awaited, **certificate of course completion** from the institute/university last studied must be provided in the prescribed format. (**Refer ANNEXURE-II**)
- iii. Bachelor’s degree certificate and marks sheet.
- iv. Certificate of Matriculation /X class as the proof of age of the candidate.
- v. Certificate of category (SC/ST/OBC-NCL/EWS), if applicable, as per Government of India format (**Refer ANNEXURE-II**), issued by the competent authority. In case of OBC-NCL/EWS category, the certificate must **be issued on or after April 01, 2025**.
- vi. Undertaking by the candidate on OBC-NCL status in the prescribed format (**Refer ANNEXURE-II**)
- vii. Certificate for Persons with Disabilities (PwD), if applicable, issued by the competent authority. (**Refer ANNEXURE-II**)
- viii. No Objection Certificate (NoC) from the Employer/Industry (**Refer ANNEXURE-II**)
- ix. Sponsorship Letter for Sponsored Ph.D. Program (**Refer ANNEXURE-II**)

Note: If the original certificates are not in English/Hindi, English/Hindi version/translation of such certificates, duly certified by the Principal/Director or other competent authority of the graduating institute, will be required during the document verification.

(Incomplete/incorrect applications without necessary supporting documents will not be considered).

12. Legal Jurisdiction

All disputes pertaining to the counselling and admission for the Ph.D. programme of NIT Goa shall fall within the jurisdiction of Goa only.

13. Important Dates

S. No.	Activity	Date
1.	Start of Online Application (Website Open)	22 May 2025
2.	Last Date for submission of Online application (Website Closure)	22 June 2025 till 5 PM 30 June 2025 till 5 PM
3.	TENTATIVE date for intimating candidates shortlisted for Written Test*/Interview	07 July 2025
3.	TENTATIVE Schedule for Written Test*/Interview	14-16 July 2025

***NOTE: Intimation regarding written test/personal interview will be posted on the Institute website. No separate interview letter/communication will be sent to individual candidates. Visit Institute website: www.nitgoa.ac.in regularly for more details and updates.**

14. Contact Details

For further queries, contact

Chairperson (Ph.D Admissions)

National Institute of Technology Goa
Cuncolim, South Goa, Goa-403703, India.

Email: phdadmissions@nitgoa.ac.in

Website: www.nitgoa.ac.in

Disclaimer

The Institute reserves the right to make changes at any time without notice, changes in and additions to the regulations, conditions governing the admission, requirements, seats, fee and any other information, or statements contained in this information brochure.

**-sd-
Chairperson
(Ph. D Admissions Cell)**